```
syms dw1 dw2 dw3 Omega I1 I2 I3

% spin about b1
I_mat = diag([I1, I2, I3]);
omega_vec = [dw1; Omega + dw2; dw3];
H_b = I_mat * omega_vec;

pretty(H_b);
```

$$\begin{pmatrix} I_1\,\mathrm{dw}_1 \\ I_2\,(\Omega+\mathrm{dw}_2) \\ I_3\,\mathrm{dw}_3 \end{pmatrix}$$

```
dw_i = I_mat \ cross(-omega_vec, H_b);
pretty(dw_i);
```

$$\begin{pmatrix} \dfrac{I_2\,\Omega\,\mathrm{dw}_3 - I_3\,\Omega\,\mathrm{dw}_3 + I_2\,\mathrm{dw}_2\,\mathrm{dw}_3 - I_3\,\mathrm{dw}_2\,\mathrm{dw}_3}{I_1} \\ -\dfrac{I_1\,\mathrm{dw}_1\,\mathrm{dw}_3 - I_3\,\mathrm{dw}_1\,\mathrm{dw}_3}{I_2} \\ \dfrac{I_1\,\Omega\,\mathrm{dw}_1 - I_2\,\Omega\,\mathrm{dw}_1 + I_1\,\mathrm{dw}_1\,\mathrm{dw}_2 - I_2\,\mathrm{dw}_1\,\mathrm{dw}_2}{I_3} \end{pmatrix}$$

```
dw_i = subs(dw_i, [dw1*dw2, dw1*dw3, dw2*dw3], [0, 0, 0]);
pretty(dw_i);
```

$$\begin{pmatrix} \dfrac{I_2\,\Omega\,\mathrm{dw}_3 - I_3\,\Omega\,\mathrm{dw}_3}{I_1} \\ 0 \\ \dfrac{I_1\,\Omega\,\mathrm{dw}_1 - I_2\,\Omega\,\mathrm{dw}_1}{I_3} \end{pmatrix}$$

```
dw_i = subs(dw_i, [I1, I2, I3, Omega], [120, 100, 80, 1])
```

```
dw_i =
```

$$\begin{pmatrix} \dfrac{\mathrm{dw}_3}{6} \\ 0 \\ \dfrac{\mathrm{dw}_1}{4} \end{pmatrix}$$

```
pretty(dw_i)
```

$$\begin{pmatrix} \frac{dw_3}{6} \\ 0 \\ \frac{dw_1}{4} \end{pmatrix}$$

```
dw_i = simplify(dw_i);
zeroRows = all(isAlways(dw_i == 0), 2);
```

```
Warning: Unable to prove 'dw1/4 == 0'.
Warning: Unable to prove 'dw3/6 == 0'.
```

```
dw_i = dw_i(~zeroRows, :);
pretty(dw_i)
```

$$\begin{pmatrix} \frac{dw_3}{6} \\ \frac{dw_1}{4} \end{pmatrix}$$

```
x2 = [dw1; dw3];
A = jacobian(dw_i, x2)
```

```
A =
```

$$\begin{pmatrix} 0 & \frac{1}{6} \\ \frac{1}{4} & 0 \end{pmatrix}$$

```
[V, D] = eigs(double(A))
```

```
V = 2×2
    0.6325   -0.6325
    0.7746    0.7746
D = 2×2
    0.2041         0
         0   -0.2041
```

```
V = V ./ V(1,1)
```

```
V = 2×2
    1.0000   -1.0000
    1.2247    1.2247
```

```
D
```

```
D = 2×2
    0.2041         0
         0   -0.2041
```